

FastCRL: A Fast Network with Adaptive Fourier Transform and Offset Prediction for Fetal Crown-Rump Length Measurement and Position Estimation in Ultrasound Images

Jiatao Liu, Ying Tan[†], Chunlian Wang, Kenli Li, Guanghua Tan, Chubo Liu

Abstract—Fetal crown-rump length (CRL) is one of the most accurate method for estimating gestational age in early pregnancy. Typically, the process of manual CRL measurement by physicians is cumbersome, prone to errors due to fetal position, and susceptible to inter-observer variability. To provide an accurate, real-time, and reliable fetal CRL measurement solution, we propose FastCRL that utilizes key landmarks detection for efficient CRL measurements and fetal position estimation. Specifically, fast and lightweight network blocks are employed for both the encoder and decoder. By outputting low-resolution heatmaps and axial offset maps of key landmarks, we achieve a balance between high accuracy and fast inference speed. A novel Lightweight Adaptive Fourier Transform (LAFT) module is introduced to globally filter noise in ultrasound images and enhance the features required for landmark prediction. Additionally, the challenge of evaluating fetal position flexion and extension is effectively addressed by analyzing the angles between key landmarks on the fetal head, buttocks, and neck. The experimental results on our dataset indicate that our method for determining fetal position is both objective and efficient. FastCRL achieves a performance level consistent with the average human expert. In terms of measuring CRL, FastCRL achieved an error rate of less than 3% in 99.1% of measurements with 32 ms latency, significantly outperforming other baselines and demonstrating substantial potential for clinical application.

Index Terms—crown-rump length measurement; deep learning; fetal position; lightweight adaptive Fourier transform; offset maps

I. INTRODUCTION

THE measurement of crown-rump length (CRL) is crucial for estimating gestational age during early pregnancy [1], and accurate dating is essential for interpreting

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The proposed model code and pre-training parameters are available at: <https://github.com/jt-liu/FastCRL>

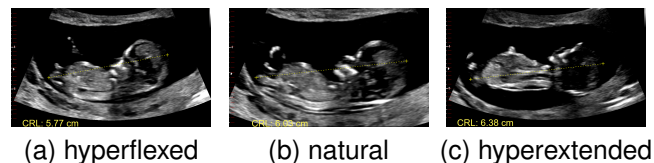


Fig. 1. Example of fetal position classification and differences in measurement values. The CRL values of a fetus in hyperflexed, natural, and hyperextended positions are 5.77cm, 6.03cm, and 6.38cm, respectively.

subsequent fetal size and growth measurements. However, manual measurement of fetal biometry from ultrasound images is time-consuming, prone to inter-observer variability, and requires expertise. More challenging, when measuring crown-rump length (CRL), as illustrated in Figure 1, fetal position—including hyperflexed, hyperextended, and natural positions—can directly influence measurement accuracy [2], [3]. Thus, there is a growing demand for automated methods to perform these tasks reliably, with recent advancements in deep learning showing promising results in automating ultrasound-based measurements. We observed the characteristics of the 3 fetal positions and found that the primary distinction among these positions lies in the relative position of the fetal neck to the head and buttocks. Estimating fetal position from this perspective presents a promising avenue for our exploration.

Deep learning-based methods have demonstrated powerful capabilities in the fields of machine cognition [4], [5] and computer vision [6], [7]. In the field of medical biometric measurement, particularly in tissue segmentation, which serves as a critical step before accurate measurements are conducted, recent research has leveraged convolutional neural networks (CNNs) to automatically segment tissues in fetal ultrasound images. This approach enables more precise biometric assessments, such as abdominal circumference [8], thalamic diameter [9], and head circumference [10], [11]. However, methods based on tissue segmentation have inherent limitations. High precision is heavily dependent on segmentation quality, and fitting errors are inevitable when selecting measurement landmarks. In addition, these two-stage methods introduce greater computational overhead, making it challenging to deploy them on edge computing devices [12], [13]. By contrast, using keypoint detection methods for end-to-end measurements can achieve more accurate performance

[14]. Furthermore, accurately capturing neutral fetal positions in ultrasound video streams for reliable CRL measurement presents significant challenges. Recently, keypoint detection algorithms have demonstrated remarkable capabilities, particularly in human pose estimation in natural images [15]–[18]. These algorithms have shown an exceptional ability to accurately identify and localize specific points of interest within complex visual data, making them highly effective in various applications.

In this paper, we present FastCRL, a novel deep learning framework designed for the rapid and accurate measurement of CRL and position estimation from fetal ultrasound images. FastCRL can simultaneously predict the head and hip measurement points as well as the neck spine point. The head and hip points are used for direct measurement of CRL, while the angle between these points and the neck spine point is utilized for estimating fetal position. The model utilizes a fast-inference backbone and integrates a lightweight adaptive Fourier transform (LAFT) module with offset map prediction within a deep neural network architecture to efficiently capture relevant anatomical features for CRL measurement and fetal position estimation. The low-latency backbone and low resolution heatmap output with offset prediction effectively improve the inference speed of the model. The LAFT module allows the network to selectively focus on relevant frequency components within ultrasound images, facilitating robust feature extraction while minimizing computational overhead. Additionally, the offset prediction further refines the measurements by incorporating spatial context, thereby enhancing the overall accuracy of the system. This approach, with an inference speed of 32 ms per frame, enables accurate, real-time measurements suitable for clinical applications.

FastCRL effectively addresses several challenges inherent in ultrasound-based CRL measurement and position estimation, including image noise, anatomical variability, and computational efficiency. In clinical applications, this is the first autonomous AI-driven reliable real-time CRL measurement method to our knowledge. The main contributions are summarized as follows: 1) We propose a novel one-time inference neural network architecture for real-time CRL measurement and fetal position estimation; 2) We introduce a lightweight global adaptive frequency domain filtering module; 3) We develop a strategy utilizing offset maps to output sub-pixel key landmark coordinates.

II. RELATED WORKS

A. Fetal CRL measurement

Crown-rump length (CRL) is significantly correlated with fetal growth parameters and serves as an indicator of the fetus's overall developmental level [19]. According to standardized guidelines [1], [20], CRL measurement is typically performed between the 9th and 13th weeks of pregnancy, using ultrasound imaging to obtain the linear distance from the top of the head to the buttocks of the fetus. Fetal position is a crucial factor influencing the accuracy of these measurements. Roux et al. [21] and Wanyonyi et al. [22] proposed examining the angles formed by fetal anatomical lines to assess fetal position. While

these methods are relatively objective, they still exhibit some inconsistency. Tan et al. [23] introduced the cranial-dorsal-hip angle as a tool for fetal position assessment. Its ability to provide a quantitative, objective measure holds significant clinical relevance. However, the method heavily relies on the accuracy of image segmentation and necessitates additional anatomical structure detectors.

B. Biometric measurement in fetal ultrasound images

In the task of fetal biometric measurement, several studies have explored the use of deep learning methods for detecting key fetal landmarks and measuring biometric parameters. For instance, Jang et al. [8] proposed a method that classifies important ultrasound features and employs Hough transformation for accurate abdominal circumference measurement. The approach addresses key challenges in obstetric diagnostics, such as low contrast, non-uniformity, and irregular abdominal shapes. Ghelich et al. [24] employed a multi-feature pyramid U-Net to segment the fetal head area, followed by the measurement of fetal head parameters like biparietal diameter and head circumference using a least squares optimization algorithm. Similarly, Plotka et al. [25] enhanced scanning plane localization by integrating attention mechanisms and stacking modules, achieving simultaneous localization, classification, and measurement of key fetal body parts. However, Gao et al. [14] highlighted inherent flaws in segmentation-based measurement methods, such as excessive reliance on segmentation results and fitting errors when selecting measurement landmarks. This indicates that single-stage end-to-end biometric measurement models have better application prospects.

Deep learning-based fetal CRL measurement methods have not yet been thoroughly explored, particularly in terms of fetal position estimation. For instance, to address the challenges that traditional CNNs face, Zhao et al. [26] integrated a deformable self-attention mechanism within a hybrid Transformer architecture, enabling effective multi-scale processing crucial for segmenting irregular fetal shapes, and then perform CRL measurement. However, this approach remains a two-stage method and is computationally expensive.

C. Keypoint detection

Keypoint detection algorithms based on deep learning have shown remarkable progress in various fields such as human pose estimation. Lifshitz et al. [27] employed CNNs to vote for the position of each keypoint in the image, leveraging information from the entire image to generate accurate keypoint predictions. Building on this, Sun et al. [15] proposed a HRNet, which maintains high-resolution representation throughout the entire process by connecting high to low-resolution sub-networks in parallel, achieving rich and detailed representations through repeated multi-scale fusion. To reduce complexity while maintaining performance, Yu et al. [16] further integrated shuffle blocks from ShuffleNet into HRNet, replacing the computationally expensive point-wise convolutions with lightweight conditional channel weighting units.

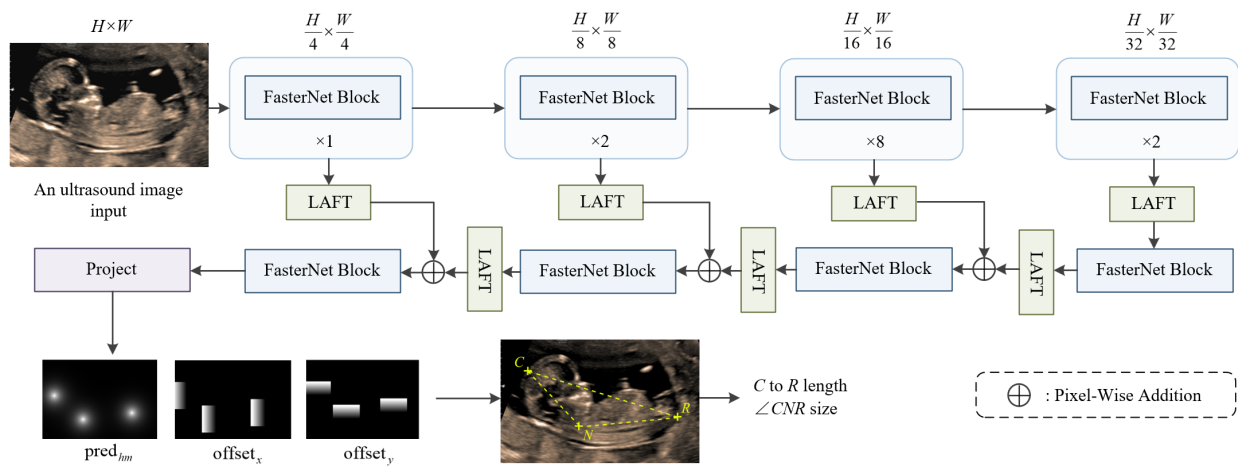


Fig. 2. Model overview. FastCRL takes a three-channel ultrasound image as input and outputs a low-resolution key landmark heatmap with two offset maps along the x and y axes. The precise key landmarks are then connected and used for fetal CRL measurement and position estimation.

In transformer-based methods, Xu et al. [18] introduced a simple yet powerful baseline model, ViTPose, that leverages plain vision transformers for pose estimation, revealing their untapped potential in this domain. Despite its design lacking specific domain knowledge, ViTPose excels in visual recognition tasks due to its simplicity, scalability, flexibility, and knowledge transferability. Yuan et al. [17] integrated the multi-resolution parallel design from HRNet and utilized local-window self-attention for efficient memory and computation. Additionally, they incorporated convolution into the feed-forward network to enable information exchange across disconnected image windows.

III. METHODOLOGY

A. Model overview

The model architecture is illustrated in Figure 2. Our model processes a three-channel ultrasound image with a resolution of $H \times W$ and generates predicted maps at a reduced resolution of $\frac{H}{4} \times \frac{W}{4}$. The deep model output consists of three components: (1) a heatmap of the key landmarks, denoted as pred_{hm} ; (2) an offset map for the predicted key landmarks along the x -axis, denoted as offset_x ; and (3) an offset map for the predicted key landmarks along the y -axis, denoted as offset_y .

Since doctors' evaluation and classification of fetal position are subjective, our method minimizes this subjectivity by leveraging automated keypoint detection to identify anatomical landmarks as reference points. Specifically, after obtaining the precise landmarks of the head (C), hip (R), and neck (N), we connect C and R to measure the CRL. We then connect N to both C and R and calculate the angle $\angle CNR$ as a reference for fetal position evaluation. This approach ensures consistent position estimation across doctors, reducing reliance on manual interpretation.

B. Feature extraction

Real-time performance is crucial for diagnosis and decision-making in ultrasound video streams. A fast-inference backbone

network can greatly shorten processing time, enabling the system to operate in real-time environments. Additionally, in remote medical services, the use of fast inference backbone networks can significantly improve user experience, reduce waiting time, and enhance interactivity.

In CNNs, spatial position information may be lost during continuous convolution and down-sampling processes. Many existing methods attempt to compensate for this loss by using dense skip connections [28], [29], but these approaches increase memory access overhead and model inference latency. To preserve positional information while meeting the requirements for CRL measurement and fetal position estimation from real-time video streams, we take FasterNet [30] as the backbone network of the model. FasterNet employs a partial convolution as its fundamental operation. As illustrated in Figure 3, a FasterNet block comprises a 3×3 partial convolution and two point-wise convolutions. The point-wise convolutions are used for channel communication. The partial convolution applies regular convolution for spatial feature extraction only on a subset of the input channels, while leaving the remaining channels unchanged. For an input tensor $\mathbb{R}^{h \times w \times c}$, where the convolution kernel size is $k \times k$ and the number of spatial feature channels extracted is c_p , the number of Flops when using partial convolution are

$$\text{Flops} = h \times w \times c_p^2 \times k^2, \quad (1)$$

the memory access amount is

$$h \times w \times 2c_p + k^2 \times c_p^2 \approx h \times w \times 2c_p. \quad (2)$$

We use a typical partial ratio of $\frac{c_p}{c} = \frac{1}{4}$ based on findings from FasterNet, where it demonstrated optimal performance. The partial ratio resulting in $\frac{1}{16}$ of the Flops required for a regular convolution and $\frac{1}{4}$ of the memory access.

C. LAFT module

Ultrasound waves typically undergo attenuation due to absorption, reflection, and scattering as they pass through tissues, resulting in ultrasound images containing substantial high-frequency noise. Additionally, unlike CRL measurement

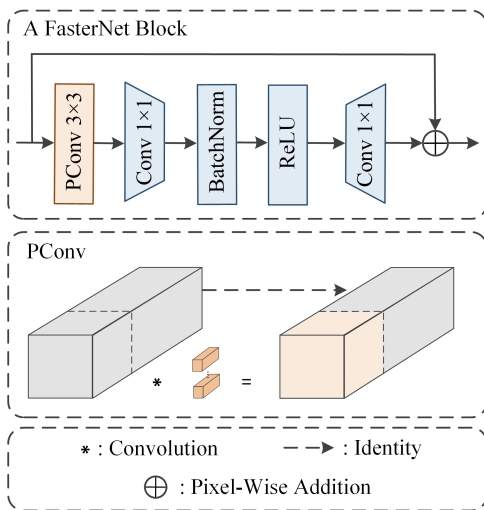


Fig. 3. A FasterNet block, which consists of a 3×3 partial convolution and two point-wise convolutions. The partial convolution operates on only a portion of the input tensor, leaving the remaining part in memory without any operation. This approach significantly reduces the number of Flops and memory access.

landmarks, a neck key landmark often lack sufficient high-frequency information. To address these challenges, we propose introducing learnable Fourier transforms at various stages of the decoder to achieve global adaptive feature filtering.

The LAFT module is designed as depicted in Figure 4, which adjusts specific frequency components in parallel at various stages of the decoder. The core concept involves applying a parameterized attention map to the feature map in Fourier space through learning. Specifically, we first map the feature maps into two independent groups, as ultrasound images contain low-frequency and high-frequency components. The low-frequency component represents the overall structure and contour of the image, while the high-frequency component captures fine details such as edges and noise. By dividing these components into two branches, LAFT allows the network to independently process each type of information, achieving specialized processing. This separation aims to mitigate the risk of feature degradation caused by treating all frequencies uniformly. Given a feature map $m \in \mathbb{R}^{h \times w \times c}$, we first apply two sets of two-layer 3×3 convolutions to split m into two distinct feature maps corresponding to high-frequency and low-frequency components, as:

$$m \xrightarrow{2 \times \text{Conv}_{3 \times 3}} (g_1, g_2) \in \mathbb{R}^{h \times w \times \frac{c}{2}}. \quad (3)$$

For each set of features, we then perform a two-dimensional Fast Fourier Transform (FFT) on the feature maps in the spatial dimension to project g into the frequency domain, as:

$$F_g = \mathcal{F}[g] \in \mathbb{C}^{h \times w \times \frac{c}{2}}, \quad (4)$$

where $\mathcal{F}[\cdot]$ represents FFT. Next, we hope that each branch uses a mechanism in the frequency domain to prioritize the most relevant features for its frequency range. Thus, we define a parameterized attention map $A_m \in \mathbb{C}^{h \times w}$ for each branch, respectively, which is then multiplied with F_g to modulate the spectrum, as:

$$F'_g = F_g \otimes A_m, \quad (5)$$

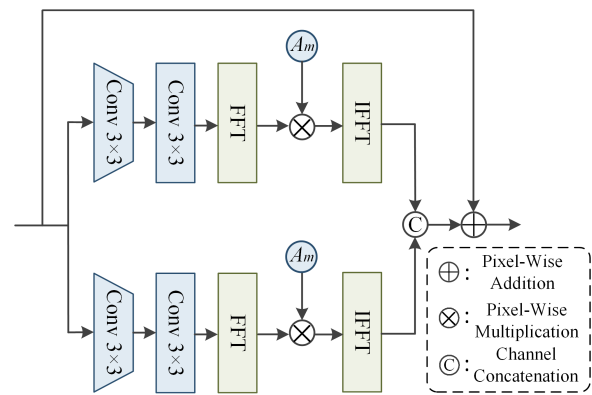


Fig. 4. Our LAFT module. Two parallel branches adaptively process high-frequency and low-frequency information, respectively.

where $\otimes$ denotes the tensor point-wise multiplication. It is important to note that we defined attention maps only in the width and height dimensions of the tensor, excluding the channel dimension, which is different from the common usage methods [31], [32]. This approach reduces the parameter count of our module by a factor of $\frac{c}{2}$. We further investigated the impact of different settings in the ablation study. Afterwards, we apply a inverse FFT to transform F'_g back into the spatial domain, as:

$$g' = \mathcal{F}^{-1}[F'_g] \in \mathbb{R}^{h \times w \times \frac{c}{2}}. \quad (6)$$

Finally, we concatenate the two sets of feature maps, g'_1 and g'_2 , along the channel dimension to obtain the feature map $m' \in \mathbb{R}^{h \times w \times c}$ after adaptive frequency filtering. This integration combines the strengths of both branches, resulting in richer feature representations. The complementary nature of the two branches ensures that neither high-frequency noise nor low-frequency distortions dominate the learning process.

D. Offset prediction

When using heatmaps to regress key landmarks, accuracy is reduced due to the rounding of pixel coordinates, a limitation exacerbated by the low resolution of the feature map. However, high-resolution feature decoding incurs significant computational costs, diminishing real-time performance. To address this, we propose a model that outputs a low-resolution heatmap while also predicting the offset of key landmarks in the width and height directions of the image, enabling sub-pixel key landmark predictions.

We define offset maps, offset_x in the width direction and offset_y in the height direction of the predicted key landmarks, to obtain sub-pixel key landmark coordinates while maintaining the speed of low-resolution image output. When loading data, since the inevitable rounding error in the heatmap after down-sampling, an offset needs to be generated within a rectangular range of $r \times r$ around the key landmark coordinates. Specifically, if the predicted key landmark offset at the coordinate position $(x_{\text{approx}}, y_{\text{approx}})$ is (O_x, O_y) , then in the offset_x , the response value at the coordinate position $(x_{\text{approx}}, y_{\text{approx}})$ is calculated as:

$$O_x = \frac{1}{r} (x - \mu x_{\text{approx}}), \quad (7)$$

correspondingly, O_y in the offset_y is calculated as:

$$O_y = \frac{1}{r} (y - \mu y_{\text{approx}}). \quad (8)$$

where (x, y) is the ground truth. During inference, after we obtained a heatmap and two offset maps, the precise key landmark coordinates $(x_{\text{accu}}, y_{\text{accu}})$ for the final prediction are calculated as:

$$(x_{\text{approx}}, y_{\text{approx}}) = \text{argmax}(pred_{\text{heatmap}}), \quad (9)$$

$$x_{\text{accu}} = \mu x_{\text{approx}} + r O_x, \quad (10)$$

$$y_{\text{accu}} = \mu y_{\text{approx}} + r O_y. \quad (11)$$

Here, the scaling factor $r = 4$ is empirically determined through a ablation study to optimize the sub-pixel accuracy, and the factor $\mu = 4$ aligns with the downsampling ratio used in the heatmap generation process, ensuring that predicted offsets correspond accurately to the original image resolution.

E. Loss function

To address the imbalance in the heatmap, where the number of positive samples is significantly smaller than that of negative samples, and to account for the consistent response area sizes in heatmaps of fetuses imaged at different ultrasound magnifications, we propose a scale-invariant focal loss function. This design is inspired by the scale-invariant loss employed in monocular depth estimation [33], [34] and the focal loss [35], [36] used in object detection. Originally, the scale-invariant loss addresses scale discrepancies by penalizing errors in logarithmic space, ensuring consistency across varying magnifications. The focal loss mitigates class imbalance by dynamically reducing the weight of easy-to-classify samples, thereby emphasizing harder-to-classify regions. By integrating the principles of these two loss functions, our approach effectively handles sample imbalance and scale variability in fetal ultrasound segmentation, improving both the robustness and precision of the model.

Our scale-invariant focal loss function consists of three components. The first is the variance of the difference between the predicted value and the ground truth:

$$L_1(h, \hat{h}) = \frac{1}{n} \sum_q \left(d_q - \frac{1}{n} \sum_p d_p \right)^2, \quad (12)$$

the second component is the pixel-level difference of negative samples:

$$L_2(h^-, \hat{h}^-) = \frac{1}{n^-} \sum_p d_p, \quad (13)$$

and the third is the pixel-level difference of positive samples:

$$L_3(h^+, \hat{h}^+) = \frac{1}{n^+} \sum_p d_p. \quad (14)$$

Here, h and $\hat{h}$ denote the predicted heatmap and the ground truth heatmap, respectively, while $\hat{h}^-$ and $\hat{h}^+$ represent the negative and positive samples in the heatmap. The variable n refers to the total number of pixels in $\hat{h}$, and for each pixel p , $d_p = h_p - \hat{h}_p$. It should be noted that positive samples

correspond to the non-zero regions in the landmark heatmap generated during training, where each landmark is modeled as a Gaussian distribution centered at its ground truth location. Non-zero pixels within this distribution are considered positive samples, indicating regions with a high probability of containing the landmark according to the model's predictions. In contrast, negative samples consist of all other pixels outside the non-zero regions, representing areas unlikely to contain landmarks. These negative samples help enhance the model's discriminative capability.

The final loss is defined as:

$$L(h, \hat{h}) = \alpha L_1 + \beta L_2 + \gamma L_3, \quad (15)$$

where balance factor $\alpha = 0.1$, $\beta = 0.1$, and $\gamma = 0.8$ are defined experimentally. This loss is applied to the heatmap and the two offset maps of the prediction, and is calculated as:

$$L(h, \hat{h}) = \sqrt{\delta L_m + \eta L_x + \sigma L_y}, \quad (16)$$

where L_m is the heatmap loss, L_x , and L_y are the offset map losses in the width and height directions of the image, respectively. Balance factor $\delta = 1.0$, $\eta = 0.1$, and $\sigma = 0.1$ are defined by experiments.

IV. EXPERIMENTS

A. Dataset

We collected 4,227 ultrasound images from 3,083 cases provided by Shenzhen Maternal and Child Health Hospital (Ethics approval number: SFYLS [2022] 068). These images were extracted from an ultrasound imaging database jointly established by Shenzhen Maternal and Child Health Hospital and 44 hospitals in China. Our dataset is sourced from Shenzhen Maternal and Child Health Hospital, Sichuan Maternal and Child Health Hospital, and Jinan Maternal and Child Health Hospital. The images were captured using seven commonly available ultrasound machines with resolutions ranging from 600×800 to 1080×1920 and were annotated by six doctors with over three years of clinical experience. The dataset was randomly divided into 2,959 images for training, 447 for validation, and 821 for testing. To verify the generalization ability of our model, we additionally extracted data from Xiamen Maternal and Child Health Hospital as an external test set, which includes 1110 images from 855 cases.

B. Implementation details

The proposed model is implemented using the PyTorch framework [37] and trained with two NVIDIA Tesla P100 GPUs. Images are uniformly down-sampled to 480×640 using the letterbox resize method. When training the model, we used AdamW [38] with a weight decay of 1×10^{-2} as the optimizer. For data augmentation, we applied random color augmentations, including adjustments to saturation, hue, brightness, and contrast. Additionally, salt and pepper noise with a ratio of 1% to 12% was added to the image as random high-frequency noise. Random Gaussian blur and motion blur have also been applied to images to simulate the artifact problem caused by the motion of ultrasound equipment probes.

For learning rate, we used the 1-cycle strategy [39] with a maximum learning rate of 3.75×10^{-4} . In the first 30% of iterations, the learning rate linearly warmed up from 1×10^{-5} to 3.75×10^{-4} , and then cosine-annealed to 1×10^{-6} . We trained for 200 epochs with a batch size of 16, taking 2.49 min per epoch. After each training epoch, validation is conducted, and the model with the highest accuracy on the validation set is saved. This model is then tested, and its performance on the test set is reported. The proposed model has approximately 33.18×10^6 parameters with 36.13×10^9 Flops.

C. Evaluation

1) *Quantitative evaluation*: Since our model is designed for measurement tasks and there are no occluded key landmark, traditional evaluation metrics for human pose estimation cannot effectively reflect the validity of the measurement model. Therefore, we propose using end point error (EPE), threshold accuracy ($\delta_i, i = 97, 98, 99$), and angle value error (AVE) to quantitatively evaluate the effectiveness of the measurement model. EPE measures the pixel error between a predicted key landmark and ground truth:

$$EPE = \frac{1}{N} \sum_k \frac{1}{b} \sum_a \sqrt{(x_{k,a} - \hat{x}_{k,a})^2 + (y_{k,a} - \hat{y}_{k,a})^2}, \quad (17)$$

δ_i measures the accuracy of predicted CRL based on a threshold i :

$$\delta_i = \frac{1}{N} \sum_m \mathbb{I} \left(|L_{\text{pred},m} - L_{\text{gt},m}| < \left(1 - \frac{i}{100} \right) \times L_{\text{gt},m} \right), \quad (18)$$

AVE measures the absolute error between the predicted fetal position angle and ground truth:

$$AVE = \frac{1}{N} \sum_m |A_{\text{pred},m} - A_{\text{gt},m}|. \quad (19)$$

Here, N represents the number of frames in the test set, b denotes the number of key landmarks in a frame, $(x_{k,a}, y_{k,a})$ is the coordinate position of key landmark a in the predicted k -th frame, and $(\hat{x}_{k,a}, \hat{y}_{k,a})$ corresponds to the ground truth. $\mathbb{I}(\cdot)$ is an indicator function that evaluates whether the error condition is met. If the condition is met, it returns 1; otherwise, it returns 0. $L_{\text{pred},m}$ is the pixel distance between the predicted head and hip key landmark coordinates in frame m , and $L_{\text{gt},m}$ is the corresponding ground truth. $A_{\text{pred},m}$ is the predicted $\angle CNR$ shown in Figure 2 to estimate fetal position in frame m , and $A_{\text{gt},m}$ is the corresponding ground truth.

We compared FasterCRL with eight state-of-the-art keypoint detection methods. To ensure a fair comparison, all models are trained using the same training strategy outlined in Section IV-B. The quantitative comparison results are presented in Table I. As shown in the table, FasterCRL achieves the highest accuracy with an inference speed of approximately 31 FPS, while utilizing 562.97 MB VRAM during inference. Additionally, we evaluated the model's inference latency on a CPU, achieving an average latency of 274 ms over 1,000 inferences on an Intel i9-12900H processor.

For fetal position estimation, we invited six doctors with 3 to 7 years of clinical experience to evaluate fetal position

in our test set and analyzed their assessments with the angle values calculated by the ground truth. As shown in Figure 5a, the angle we proposed for estimating fetal position effectively reflects the actual fetal positioning. However, only 436 out of 821 images are judged to be in the same position by all six doctors. Thus, determining the position of the fetus is still a very subjective matter, and different observers are likely to have different judgments, especially for slightly flexed positions.

To standardize the evaluation, two angles ($angle_1 = 134.20^\circ$, $angle_2 = 169.42^\circ$) are calculated based on the assessments of the six doctors to minimize errors. These angles are used to classify fetal positions as hyperflexed, neutral, or hyperextended. Specifically, if $\angle CNR$ is less than $angle_1$, the fetal position is classified as hyperflexed, and if it is greater than $angle_2$, the fetal position is classified as hyperextended. This classification method achieved an accuracy of 83.15% (830 misclassifications in 821×6 judgments), providing meaningful clinical reference value. The individual accuracies of the six doctors are 82.70%, 88.19%, 76.37%, 80.51%, 87.82%, and 83.31%, with a 95% confidence interval for the average accuracy of [78.45%, 87.85%]. Using this classification system, we evaluated the accuracy of FastCRL in determining fetal position (Figure 5b) and achieved an accuracy of 83.21% (827 misclassifications in 821×6 judgments). This result falls within the experts' confidence interval, indicating no statistically significant difference between the model and the experts' judgments.

To evaluate the generalization capability of the model, we conducted external testing using 1,110 images from Xiamen Maternal and Child Health Hospital. An expert doctor with nearly 10 years of clinical experience was invited to assess the model's measurement and pose estimation predictions. The evaluation results indicate that 14 measurements contained errors, yielding a measurement accuracy of 98.74%. The confusion matrix comparing fetal position between manual evaluation and FastCRL predictions is presented in Table II. The model achieved an accuracy of approximately 83.60% in determining fetal position, demonstrating that FastCRL is comparable to the average judgment accuracy of human evaluators. Moreover, prediction errors are primarily concentrated in cases where the model classified the position as natural despite manual judgment identifying it as hyperflexed. This indicates that effectively defining and distinguishing slightly flexed positions remains a challenging task.

2) *Qualitative evaluation*: We choose LiteHRNet-18 [16] and ViTPose-s [18] as our main competitors based on accuracy and inference speed. As shown in Figure 6, FastCRL demonstrates more accurate key landmark prediction performance, whether locating key landmarks from high-frequency or low-frequency information.

Among the 1,110 external test images, 14 were identified by experts as containing measurement errors. We analyzed these erroneous cases and identified three primary causes of measurement errors, as illustrated in the Figure 7. First, incomplete positioning occurred at the boundaries of the fetal imaging. Second, the presence of nearby tissues at the measurement point led to positioning inaccuracies. Third, the

TABLE I
QUANTITATIVE COMPARISON WITH OTHER KEYPOINT DETECTION METHODS. FLOPS ARE CALCULATED WITH AN INPUT RESOLUTION OF 480×640 .
BOLD: BEST; UNDERSCORE: SECOND BEST.

Methods	Param.(M)	Flops(G)	EPE	δ_{97}	δ_{98}	δ_{99}	AVE	Latency(ms)
HRNet-32 [15]	28.54	48.26	4.00	0.984	0.934	0.715	1.91	91
HRNet-48 [15]	63.60	98.72	4.05	0.983	0.934	<u>0.726</u>	1.94	123
LiteHRNet-18 [16]	1.13	<u>1.73</u>	<u>3.80</u>	<u>0.989</u>	<u>0.944</u>	0.689	<u>1.70</u>	33
SCNet-50 [40]	23.51	24.39	3.91	0.985	0.927	0.680	1.80	38
RSN-18 [41]	9.02	13.42	4.94	0.945	0.845	0.537	2.18	38
HRFormer-s [17]	7.75	18.73	4.75	0.976	0.888	0.629	2.32	91
ViTPose-s [18]	24.67	8.93	4.42	0.972	0.914	0.631	2.25	20
PVTv2-B2 [42]	<u>1.96</u>	1.55	4.62	0.976	0.892	0.607	2.01	41
FastCRL	33.18	36.13	3.46	0.991	0.960	0.784	1.54	<u>32</u>

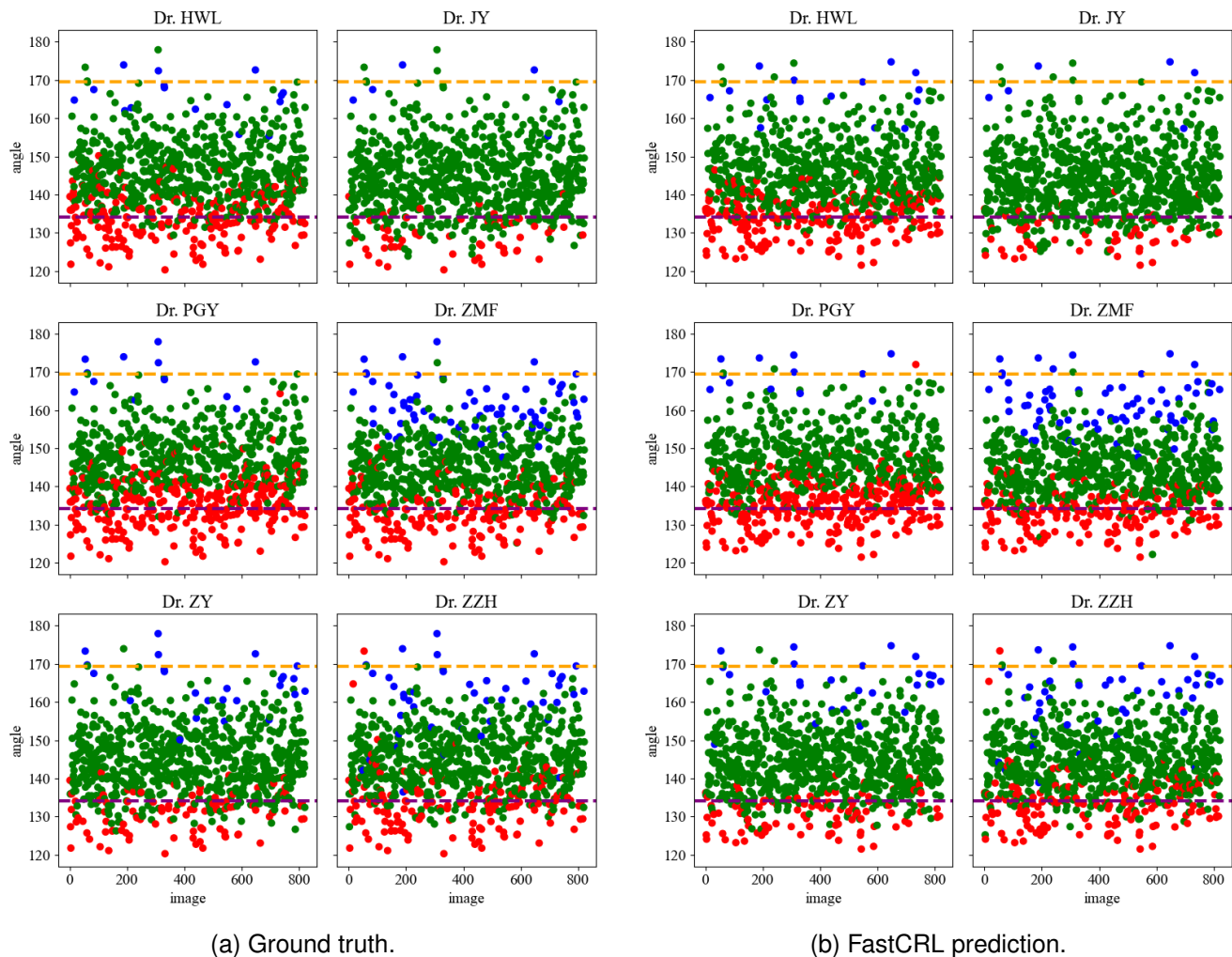


Fig. 5. Visualization of fetal position determined by six doctors in the test set combined with $\angle CNR$ size. The position of a fetus marked by the doctor as hyperextended is represented by a blue dot, as natural by a green dot, and as hyperflexed by a red dot. The two angles used to divide fetal position types are $angle_1 = 134.20^\circ$, represented by a purple dashed line, and $angle_2 = 169.42^\circ$, represented by an orange dashed line.

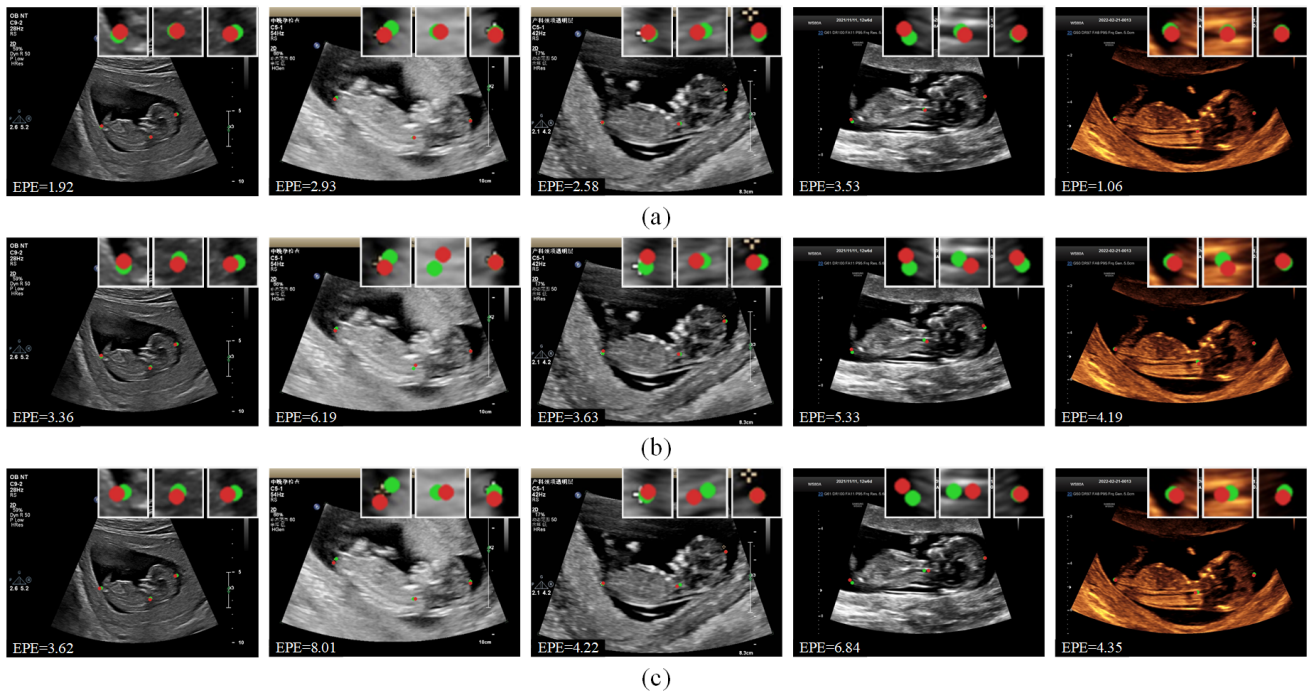


Fig. 6. Qualitative Evaluation. Comparison of (a) FastCRL with (b) LiteHRNet-18 and (c) ViTPose-s, where the green dots represent the ground truth and the red dots represent key landmarks predicted by the model. The prediction of enlarged key landmarks is displayed above each image in the order of hip, neck, and head landmarks.

TABLE II
THE CONFUSION MATRIX COMPARING FETAL POSITION IN EXTERNAL TEST SET BETWEEN MANUAL EVALUATION AND MODEL PREDICTIONS.

Model \ Doctor	Hyperflexed	Natural	Hyperextended
Hyperflexed	117	7	0
Natural	159	793	9
Hyperextended	0	7	18

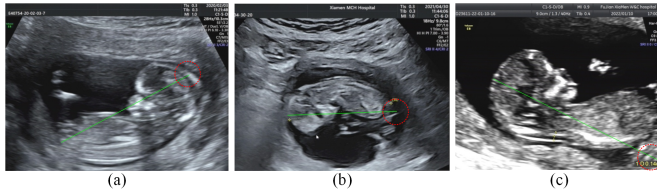


Fig. 7. Three primary causes of measurement errors identified by human experts in the external test set: (a) incomplete positioning at the boundaries of the fetal imaging, (b) positioning inaccuracies caused by nearby tissues at the measurement point, and (c) measurement points extending beyond the imaging area.

measurement point extended beyond the imaging area.

To analyze feature changes before and after adaptive frequency filtering, we visualized the feature maps before and after FFT and IFFT operations in LAFT module, as shown in Figure 8. The visualizations reveal that our global adaptive filtering significantly alters feature representations, with each filtering branch exhibiting distinct focuses. For the encoder features, one branch preserves high-frequency textures and spatial position details, while the other emphasizes the fetus's overall position in the low-frequency domain; For the decoder

features, one branch activates the surrounding regions of key landmark positions to guide initial decoding, whereas the other normalizes the key landmark positions to improve the accuracy of coordinate localization during decoding.

To further assess the impact of the two parallel branches in the LAFT module, we visualized the activated features of both branches using Grad-CAM [43] at feature scales of $\frac{H}{4} \times \frac{W}{4}$ and $\frac{H}{32} \times \frac{W}{32}$, respectively. As shown in Figure 9, the figure illustrates the distinct emphasis of each branch. It should be noted that the visualizations are generated without specifying a specific output class, as the output is continuous and not explicitly classified. At a feature scale of $\frac{H}{32} \times \frac{W}{32}$, one branch focuses on global low-frequency regions, such as areas outside the imaging sector, while another branch concentrates on locating the fetus within high-frequency regions. At a feature scale of $\frac{H}{4} \times \frac{W}{4}$, one branch accurately targets the head and hip measurement landmarks in high-frequency regions, while another branch enhances the response to low-frequency neck regions rather than focusing on the precise positioning of the head and hip measurement landmarks.

D. Ablation study

1) *Model structure*: We first conduct ablation experiments on the LAFT module and offset prediction in FasterCRL. We removed the LAFT module and offset prediction separately and trained FastCRL without these components. The quantitative comparison results are presented in the Table. As shown in Table III, the incorporation of the two proposed modules significantly improves the accuracy of key landmarks prediction for FastCRL. Additionally, we observed that the

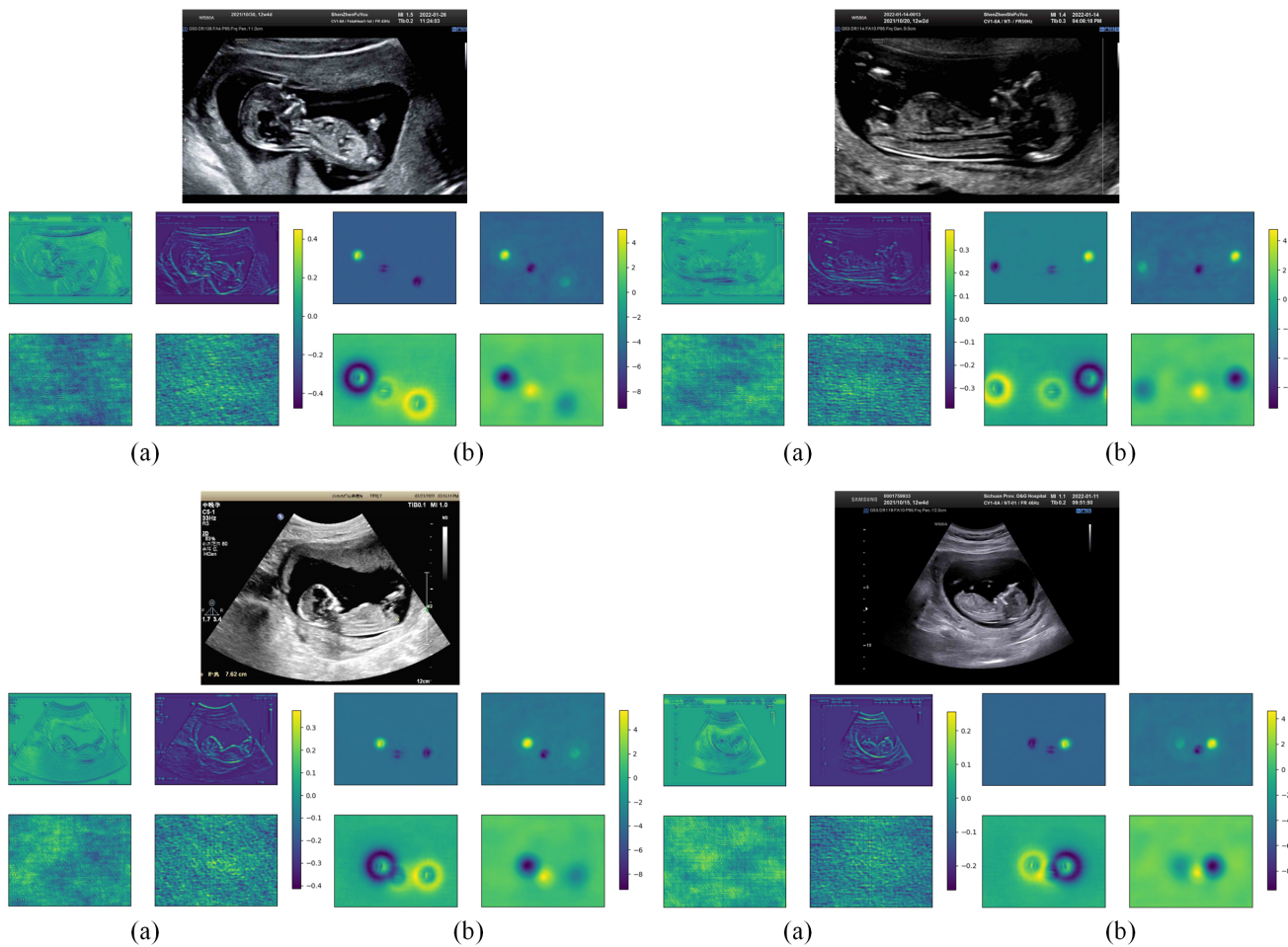


Fig. 8. Visualization of feature maps before and after FFT and IFFT operations in the LAFT module. For each input image, the feature changes resulting from the LAFT module's application in (a) the encoder and (b) the decoder are presented, highlighting differences before and after the adaptive global filtering operation. In each set of four feature maps, the first and second rows depict features before and after filtering, respectively, while the first and second columns illustrate the feature changes in the two parallel branches. The feature map shown in the figure is the average of multi-channel feature maps.

TABLE III
ABLATION STUDY ON THE MODEL STRUCTURE.

LAFT	offset	EPE	δ_{97}	δ_{98}	δ_{99}	AVE
$\times$	$\times$	4.41	0.974	0.921	0.691	2.09
$\checkmark$	$\times$	4.15	0.985	0.938	0.719	1.99
$\times$	$\checkmark$	3.70	0.978	0.939	0.748	1.61
$\checkmark$	$\checkmark$	3.46	0.991	0.960	0.784	1.54

model using only the LAFT module achieved a higher δ_{97} than the model using only offset prediction, but other indicators showed degradation. We speculate that it is because the LAFT module emphasizes coarse positioning, while the offset prediction module focuses on fine-tuning the predicted landmarks within a smaller range. The combination of these two modules allows the model to achieve optimal performance.

To validate the effectiveness of our LAFT module in capturing global frequency-domain attention, we replaced the FFT and IFFT operations in the LAFT module with traditional attention mechanisms, specifically SE [44] and CBAM [45]. We then trained the modified models for comparison. As shown in Table IV, the results demonstrate the superior performance of our LAFT module, which indicates that global frequency

TABLE IV
QUANTITATIVE COMPARISON BETWEEN OUR LAFT MODULE AND TWO TRADITIONAL ATTENTION MECHANISMS.

module	parallel	EPE	δ_{97}	δ_{98}	δ_{99}	AVE
SE	$\times$	4.48	0.955	0.894	0.680	2.09
SE	$\checkmark$	4.44	0.949	0.878	0.654	2.03
CBAM	$\times$	4.37	0.968	0.918	0.681	1.94
CBAM	$\checkmark$	4.32	0.968	0.893	0.680	1.92
LAFT	$\checkmark$	3.46	0.991	0.960	0.784	1.54

processing can effectively improve the model's attention to important information for ultrasound images.

We further investigate the impact of settings in the LAFT module on model performance. As shown in Table V, we conducted ablation experiments on the shape of the parameterized attention map A_m and the parallel structure of the two branches. The results demonstrate that using attention maps with $\frac{c}{2}$ times fewer parameters, combined with the parallel architecture of two branches, significantly improves model performance. It is worth noting that the parallel architecture significantly enhances network performance for an attention map with a shape of $h \times w$ but is not effective for $h \times w \times c$. As

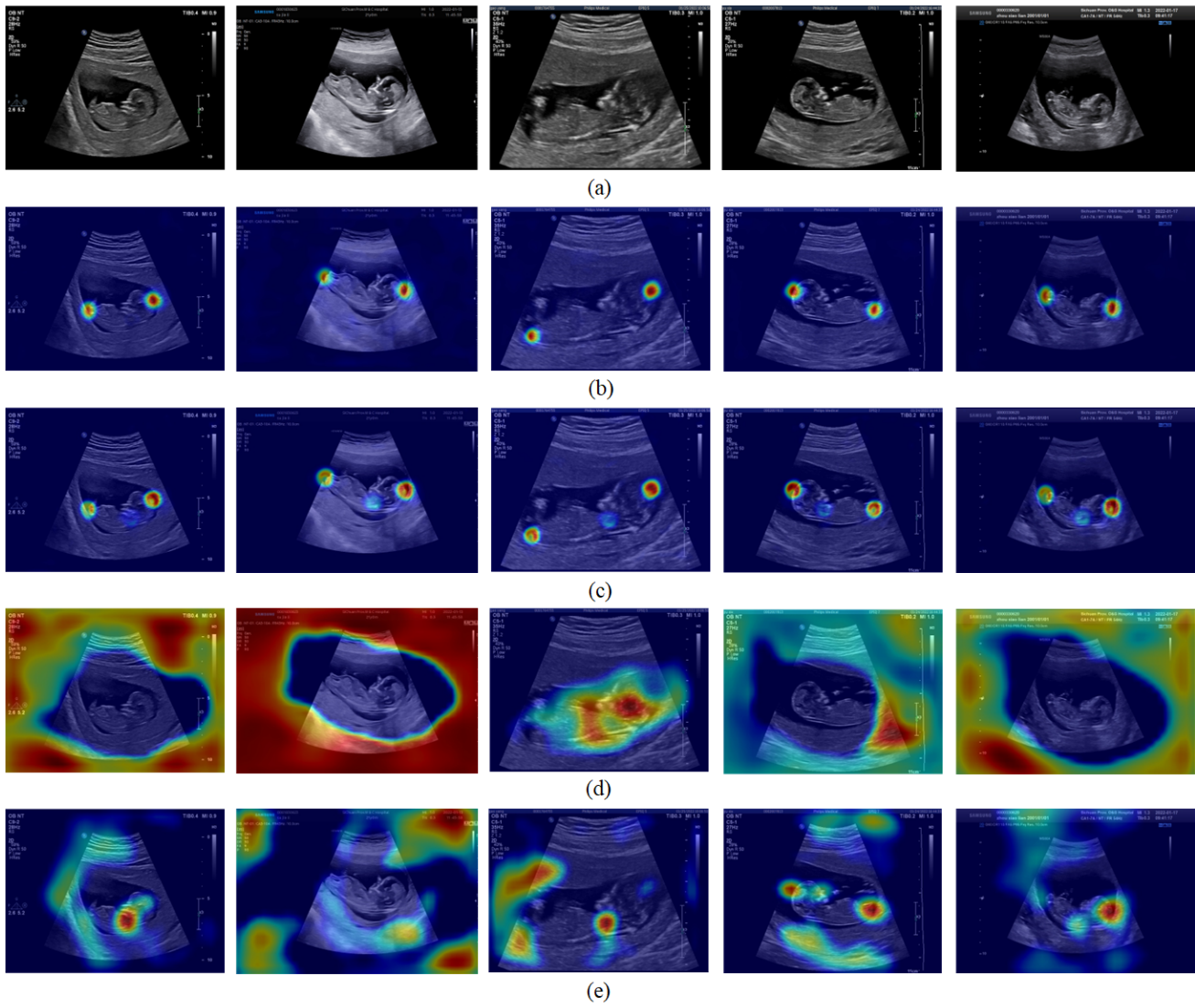


Fig. 9. Visualization of feature activations obtained from different branches in our LAFT module at two stages. (a) Input images. Visualization of feature activations for the two branches when the feature scales are (b) (c) $\frac{H}{4} \times \frac{W}{4}$ and (d) (e) $\frac{H}{32} \times \frac{W}{32}$, where warmer colors indicate the spatial regions with stronger activation in the two branches.

TABLE V
ABLATION STUDY ON THE SETTINGS OF OUR LAFT MODULE.

A_m	parallel	EPE	δ_{97}	δ_{98}	δ_{99}	AVE
$h \times w \times c$	✗	4.72	0.984	0.929	0.683	1.82
$h \times w \times c$	✓	4.94	0.981	0.915	0.671	1.80
$h \times w$	✗	4.95	0.978	0.914	0.677	1.91
$h \times w$	✓	3.46	0.991	0.960	0.784	1.54

TABLE VI
ABLATION STUDY ON THE RECTANGULAR RANGE r IN THE OFFSET MAP.

r	EPE	δ_{97}	δ_{98}	δ_{99}	AVE
48	3.59	0.989	0.955	0.769	1.63
36	3.56	0.990	0.954	0.771	1.58
24	3.52	0.991	0.955	0.777	1.54
12	3.48	0.990	0.959	0.775	1.55
4	3.46	0.991	0.960	0.784	1.54

discussed in Section IV-C2, incorporating global frequency-domain attention within a parallel structure improves the adaptive filtering capabilities for high and low frequency information, respectively. However, extending the mechanism to include the channel dimension increases complexity, which can dilute the benefits of the parallel architecture. Employing a parallel strategy in this context may introduce channel attention redundancy and interference in the frequency domain space, thereby diminishing performance gains.

2) *Rectangular range in offset prediction:* In the offset map, the range of generated offsets can impact the model's accuracy. Therefore, we experimented with different rectangular ranges, r , to generate offset map labels and trained the model accordingly. As shown in Table VI, training the model with a smaller r results in higher accuracy. This suggests that the model struggles to learn large-scale offsets, aligning with our objective of fine-tuning predictions within a small range of offsets.

TABLE VII
ABLATION STUDY ON THE LOSS FUNCTION.

loss function	EPE	δ_{97}	δ_{98}	δ_{99}	AVE
MSE loss	3.59	0.990	0.945	0.773	1.66
Ours	3.46	0.991	0.960	0.784	1.54

3) *Loss function*: To verify the impact of our scale-invariant focus loss on model performance, we trained FastCRL using the commonly used mean squared error (MSE) loss and reported its performance, as shown in Table VII. It can be observed that our loss function enables the model to perform better in small-scale fine-tuning, thereby improving overall model performance.

V. CONCLUSION

We proposed an automated real-time method for fetal CRL measurement and position estimation from ultrasound images. The proposed approach addresses several challenges inherent in traditional CRL measurement practices, including the inefficiency of manual measurements and the variability due to different fetal positions. By employing a fast backbone network and a novel offset prediction strategy, FastCRL maintains high inference speed without compromising the accuracy of sub-pixel key landmark localization. The LAFT module effectively filters out noise in ultrasound images and enhances low-frequency information in the fetal neck region, ensuring reliable CRL measurements even in the presence of high-frequency interference. The experimental results demonstrate that FastCRL improves both measurement accuracy and the robustness of fetal position estimation. By significantly reducing operator dependency and automating measurement processes, our method has the potential to streamline clinical workflows, enhance measurement consistency across practitioners, and reduce healthcare professionals' workload. Furthermore, FastCRL represents an advancement in obstetric diagnostics, offering the potential to improve patient care quality and accessibility through efficient and accurate fetal biometry.

Despite the model has demonstrated improvements in fetal position estimation, clinical applications may still exhibit variability due to subjective factors, such as clinicians' classification of fetal position. Potential future improvements may incorporate contrastive learning to enhance the model's ability to distinguish subtle anatomical differences and employ advanced data generation techniques to enrich the training data with diverse fetal postures. Additionally, as our model requires a GPU-equipped device for real-time performance, further investigation into its application on standard portable ultrasound devices is needed. Techniques such as knowledge distillation and other model compression methods may offer viable solutions.

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